Roll No................

End Semester Examination

Name of the Program: B.Tech(ECE)
Name of the Course: Wireless Communication

Semester: VI
Course Code: TEC 601

Time: 3 Hours

Maximum Marks: 100

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty
(iv) Each question carries 10 marks.

Q1	(20 marks)	**CO1**
(a)	(i) Calculate the area of a hexagonal cell of radius R. (ii) Explain Hard, Soft and Softer hand-offs.	
(b)	Draw the seven cell reuse pattern and find the co-channel reuse ratio in terms cluster size N.	
(c)	How cell splitting is performed? Why it can increase the system capacity?	
Q2	(20 marks)	**C02**
(a)	State significance of two-ray ground reflection model. Explain using suitable diagrams and obtain an expression for the received signal power.	
(b)	(i) Assuming 7 cell cluster and UE at the cell boundary, find an expression to indicate SIR in co-channel cells in the downlink. (ii) A cellular system has a total of 32 cells, with radius of 1.6 km for each cell, using 7 cells in a cluster. It has been assigned a total of 336 duplex channels to handle the voice traffic. What is the area covered by the cellular system? Find the number of channels per cell and state how many simultaneous calls are possible over the cellular system.	
(c)	What do you mean by distorted transmission? Explain the term constructive and destructive interference.	
Q3	(20 marks)	**CO3**
(a)	Explain different types of small-scale fading using suitable diagrams.	
(b)	Assume mobile equipment operates over 900 MHz frequency and moves with a velocity of 96 kmph. Find the data rate to receive distortion less transmission.	
(c)	What do you mean by power delay profile? Sketch typical power delay profile to explain different delay parameters of wireless mobile environment.	
Q4	(20 marks)	**CO4**
(a)	How does Selection Combining function? Derive an expression to represent mean SNR improvement realized in Selection Combining.	
(b)	What do you mean by Rayleigh fading Model? Derive an expression to represent PDF of SNR under Rayleigh fading Model.	
(c)	Consider a transmitter operating at 1800 MHz transmits 4 watt power. Assume path loss exponent to be 4, shadow effect of 10.5 dB and the power at reference point (d_0= 100m) is - 32 dB. Find the received power at a distance of 3 km from transmitter and the allowable path loss.	

Q5	(20 marks)	
(a)	State and explain the properties of maximum length sequence?	**CO5 & CO6**
(b)	Discuss the processes involved in direct sequence spread spectrum (DSSS) and frequency hop spread spectrum (FHSS).	
(c)	Explain CDMA and discuss its salient features.	

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END SEMESTER EXAMINATION - 2024

Name of the Course: **B.Tech(ECE)** Semester: **VI**
Name of the Paper: **Microwave Engineering** PaperCode:**TEC602**

Time: 3Hours Maximum Marks: 100

Note:-
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20 marks)	
(a)	**Explain TM modes propagation in a circular wave guide.**	**CO1**
(b)	**Show that for a TE_{10} mode, a frequency of 8 GHz will pass through a wave-guide of dimensions a=1.5 cm and b=1 cm, if a dielectric with $\varepsilon_r = 4$ is inserted inside the guide.**	
(c)	**A hollow rectangular wave guide has dimensions a=4 cm and b=2 cm. Calculate the amount of attenuation if the frequency is 3 GHz.**	
Q2	(20 marks)	
(a)	**Draw the schematic of a reflex klystron and explain its working principle.**	**CO3**
(b)	**Explain tunneling when different bias potentials are applied to a tunnel diode. Show the negative resistance region and discuss its significance to design microwave sources.**	
(c)	**Discuss the working of IMPATT and TRAPATT diode for microwave signal amplification.**	
Q3	(20 marks)	
(a)	**Draw and explain Magic TEE. Write down its S-matrix and all the properties.**	**CO2**
(b)	**Explain the need of a power divider and combiner? Prove that a lossless, matched and reciprocal 3-port device is not feasible all the time.**	
(c)	**Write scattering matrix of ideal isolator. In an isolator the isolation is 15dB and insertion loss is 2dB. Find its scattering parameter.**	
Q4	(20 marks)	
(a)	**Explain the mechanism of microwave frequency and VSWR measurement.**	**CO4**
(b)	**A 90W power source is connected to the input of directional coupler with C= 20dB, D=35dB and insertion loss of 0.5dB. Find the output powers at the through, coupled andisolated ports. Assume all ports to be matched.**	

(c)	**Explain network analyzer with the help of a block diagram.**	
Q5	(20 marks)	CO5, CO6
(a)	**Discuss Impedance and Frequency Scaling technique used for filter transformation.**	
(b)	**Explain filter implementation technique using Richard transformation.**	
(c)	**Design a low-pass filter for fabrication using microstrip lines. The specifications include a cutoff frequency of 4 GHz, an impedance of 50 Ω, and a third-order 3 dB equal-ripple passband response.**	

3.0 dB Ripple

N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
1	1.9953	1.0000									
2	3.1013	0.5339	5.8095								
3	3.3487	0.7117	3.3487	1.0000							
4	3.4389	0.7483	4.3471	0.5920	5.8095						
5	3.4817	0.7618	4.5381	0.7618	3.4817	1.0000					
6	3.5045	0.7685	4.6061	0.7929	4.4641	0.6033	5.8095				
7	3.5182	0.7723	4.6386	0.8039	4.6386	0.7723	3.5182	1.0000			
8	3.5277	0.7745	4.6575	0.8089	4.6990	0.8018	4.4990	0.6073	5.8095		
9	3.5340	0.7760	4.6692	0.8118	4.7272	0.8118	4.6692	0.7760	3.5340	1.0000	
10	3.5384	0.7771	4.6768	0.8136	4.7425	0.8164	4.7260	0.8051	4.5142	0.6091	5.8095

Source: Reprinted from G. L. Matthaei, L. Young, and E. M. T. Jones, *Microwave Filters, Impedance-Matching Networks, and Coupling Structures*, Artech House, Dedham, Mass.,1980, with permission.

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Name of the course: B. Tech (ECE) *Semester: VI*

Name of the Paper: VLSI Technology and Design *Paper Code:* **TEC 603**

Time: 3 Hours Maximum Marks: 100

Note:

(i) All questions are compulsory
(ii) Answer any two **questions** among a, b, & c ***in each main question***.
(iii) Total marks in each main question are twenty
(iv) Each question carries 10 marks

Q1	(20 marks)	**CO1**
(a)	What is Integrated Circuit? Explain advantages of ICs.	
(b)	What do you understand by Metallurgical grade silicon (MGS), explain the process of transformation of MGS into Electronic Grade Silicon (EGS).	
(c)	Explain Doping and auto-doping, Buried layers.	
Q2	(20 marks)	**CO2**
(a)	Explain wet etching and Dry etching process in VLSI.	
(b)	Explain chemical vapor deposition technique for metallization applications in VLSI.	
(c)	Explain in brief about types of photoresists and its applications in VLSI.	
Q3	(20 marks)	**CO3**
(a)	Consider an n – channel metal oxide semiconductor field effect transistor (MOSFET) with gate to source voltage of 1.5V. Assume that W/L = 4, $\mu n C_{OX}$ = 60 × 10^{-6} AV^{-2}, and the threshold voltage is 0.6V. In the saturation region, find the drain conduction g_d (in micro Siemens).	
(b)	Derive the expression for current vs voltage for MOS in non-saturation and saturation mode.	
(c)	The slope of the $ID\ vs\ .\ VGS$ curve of an n – channel MOSFET in linear region is $10^{-4}\ \Omega^{-1}$ at $V_{DS} = 0.2V$. For the same device, neglecting channel length modulation, the slope of the $\sqrt{ID}\ vs\ VGS$ curve ($in\ \sqrt{A/V}$) under saturation region is approximately.	
Q4	(20 marks)	**CO4**
(a)	Explain in detail about resistive load MOS inverter.	
(b)	Explain briefly about switching power Dissipation in CMOS inverter.	
(c)	Find the gain of active load MOSFET amplifier.	
Q5	(20 marks)	**CO5/6**
(a)	Draw the stick diagram of expression {AB+E+CD}'	
(b)	Draw step by step layout for complement of {(A+B)CD}.	
(c)	What are the various industrial applications of CMOS? Explain in brief.	

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Name of the Program: BTech(ECE) **Semester: VI**
Name of the Course: OOPs Using C++ **Course Code: TOE-610**

Time: 3 Hours Maximum **Marks: 100**

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Explain how inheritance, function overriding, and abstract classes are used to support run-time polymorphism.	(CO1,CO2, CO4)
(b)	Write a C++ program to create a class called LogicTest with following members: Private Data Member: string nstr Public Function Members: void read(string) ➔ to accept a sentence. void evenodd() ➔ Counts number of vowels in a sentence and then check if count is even or odd. In main function create object of class LogicTest and perform the above task.	
(c)	Differentiate between a constructor and a destructor. Write a program that demonstrates the use of constructor overloading.	
Q2	(20 marks)	
(a)	What is operator overloading in C++? Write a C++ program to overload binary + operator without using friend function.	(CO1,CO3, CO4)
(b)	Describe virtual base class in C++? What is diamond problem in C++? Write a C++ program to resolve diamond problem.	
(c)	Write a C++ file handling program to read only alphabet from a given file and then count number of alphabet in a given file.	
Q3	(20 marks)	
	Explain following with help of C++ program.	

(a)	(i) Static members (ii) Scope Resolution Operator	
(b)	What are function templates? Write a C++ program to add two variables of any type using function templates.	**(CO1,CO4, CO6)**
(c)	When ambiguity error were arise in multiple inheritance, provide the Solution to the error with a C++ code.	
Q4	(20 marks)	
(a)	How is run-time polymorphism achieved in C++? Explain pure virtual function and abstract class in C++.	**(CO1,CO2, CO4)**
(b)	Write a C++ program to merge two integer arrays into third array. After merging count and print number of palindrome in a third array.	
(c)	Create an abstract class VirtualDemo with following pure virtual function: virtual void prime(int)=0; Create a derived class CalPrime which inherit abstract class VirualDemo and override the function. void prime(int) method will check whether passing number is prime or not.	
Q5	(20 marks)	
(a)	What is use of try, catch and throw keyword in Exception Handling. Write a C++ program that throws an exception of index out of bound.	**(CO1,CO2, CO5)**
(b)	Write a C++ program that demonstrate working of constructor and destructor in base and derived class.	
(c)	Explain following object oriented concept with the help of C++ program. (i) Encapsulation (ii) Inheritance (iii) Polymorphism	

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End Semester Examination 2024

Name of the Course: B.Tech-ECE | Semester: 6th

Paper Name: Advanced Embedded System | Paper Code: **TEC 659**

Time: 3 Hours | Maximum Marks: 100

Note:
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	**20 Marks**	
(a)	Explain how we can design a currency counter? Explain with the help of block diagram and flowchart.	CO1
(b)	What are Quality Attributes of Embedded system?	CO1
(c)	An embedded system-based visitor counter is needed. The system must be designed such that it can count only upto 100 visitors and after than it must blink the RED led. Explain the Embedded System design process for such system with the help of proper block diagram, flow chart for algorithm etc.	CO1
Q2	**20 Marks**	
(a)	Explain the PIC microcontroller with the help of its features and architectures.	CO2
(b)	What is the difference between Jump and Call instructions? How they get executed explain in details.	CO2
(c)	Explain SFRs in details. How can we program ports of 8051 using SFRs?	CO2
Q3	**20 Marks**	
(a)	Explain Load Sensor (Analog) interfacing with 8051. Also explain the code required in support of your answer.	CO3
(b)	A LED 7-segment display is connected to P0. Write a program to count from 0 to 9 with specific interval of 1 sec.	CO6
(c)	Explain Programming framework in Embedded Systems design.	CO3
Q4	**20 Marks**	
(a)	Explain Embedded System Design.	CO4
(b)	What is the role of Real time operating systems (RTOS) in Embedded system design?	CO4
(c)	What is Task scheduling?	CO4
Q5	**20 Marks**	
(a)	An automobile industry requires to open the AIR bags at high impact. Explain which type of microcontroller can be used to monitor such action. Which type of microcontroller unit can be used for such action among 8051, AVR, ARM 32? Make a block diagram and flow chart in support of your answer.	CO5
(b)	Draw the Architecture of 8096 and explains it's blocks in details.	CO5
(c)	What is AMBA bus in ARM 32 microprocessor? Explain in detail with the help of diagram.	CO6